

1. Purpose of this Lab

Each group selects ONE information system and carries it through the complete Systems Development Life Cycle (SDLC) — from requirement analysis to a basic prototype and testing plan.

2. Required Sections in the Project Report

Every submitted project must contain the following sections, in this order:

2.1 Project Proposal / Introduction

- Name and objective of the chosen system
- Scope and limitations
- Problem statement (what is wrong with the existing/manual system)

2.2 Requirement Analysis

- Functional requirements (what the system must do)
- Non-functional requirements (performance, security, usability)
- Data collection method used (mock interview / questionnaire, written up even if simulated)

2.3 Feasibility Study

- Technical feasibility
- Economic feasibility
- Operational feasibility

2.4 Structured Analysis Tools

- Context Diagram (Level-0 DFD)
- DFD Level-1 and Level-2 (as needed)
- ER Diagram (Entity Relationship Diagram)
- Data Dictionary

2.5 System Design

- Database design: normalization (1NF $\rightarrow$ 3NF), table schema, primary/foreign keys
- Input design: data-entry form layouts
- Output design: report / receipt / list formats

- File organization method used (sequential / indexed / direct access)

2.6 Prototype / Basic UI

- A simple front-end or wireframe/mockup of key screens

2.7 Testing Plan

- List of test cases
- Unit testing vs. system testing approach

2.8 Implementation Plan

- Hardware/software requirements
- Project schedule (Gantt chart)
- Security and disaster-recovery considerations (brief)

2.9 Conclusion and Ethics Note

- Key lessons learned by applying the SDLC
- Ethical considerations (e.g., data privacy, accuracy, access control)

3. Recommended Tools

- Diagramming: Draw.io, or Visual Paradigm (for DFD/ER diagrams)
- Database: MS Access, MySQL, or Oracle (for schema and queries)
- Scheduling: MS Project or a simple Gantt chart tool
- Prototype: MS Access forms, or a basic HTML/PHP front-end

4. Suggested Project Topics ()

No.	Suggested System
1	Library Management System
2	Inventory / Warehouse Management System
3	Voter Identity Management System
4	Payroll Management System
5	Hospital / Clinic Management System
6	Student Result Management System
7	Hotel Reservation System
8	Online Bus / Train Ticket Booking System
9	Vehicle Rental / Garage Management System
10	Employee Attendance & Leave Management System
11	Blood Bank Management System

12	Restaurant Order Management System
13	Courier / Parcel Delivery Tracking System
14	Pharmacy Management System
15	Bank Account Management System (simplified)

5. Submission Guidelines

- Submit as a single report (Word/PDF) containing all sections above, in order
- Diagrams should be neatly drawn/exported, not hand-scanned unless legible
- Group size: 2 students per group